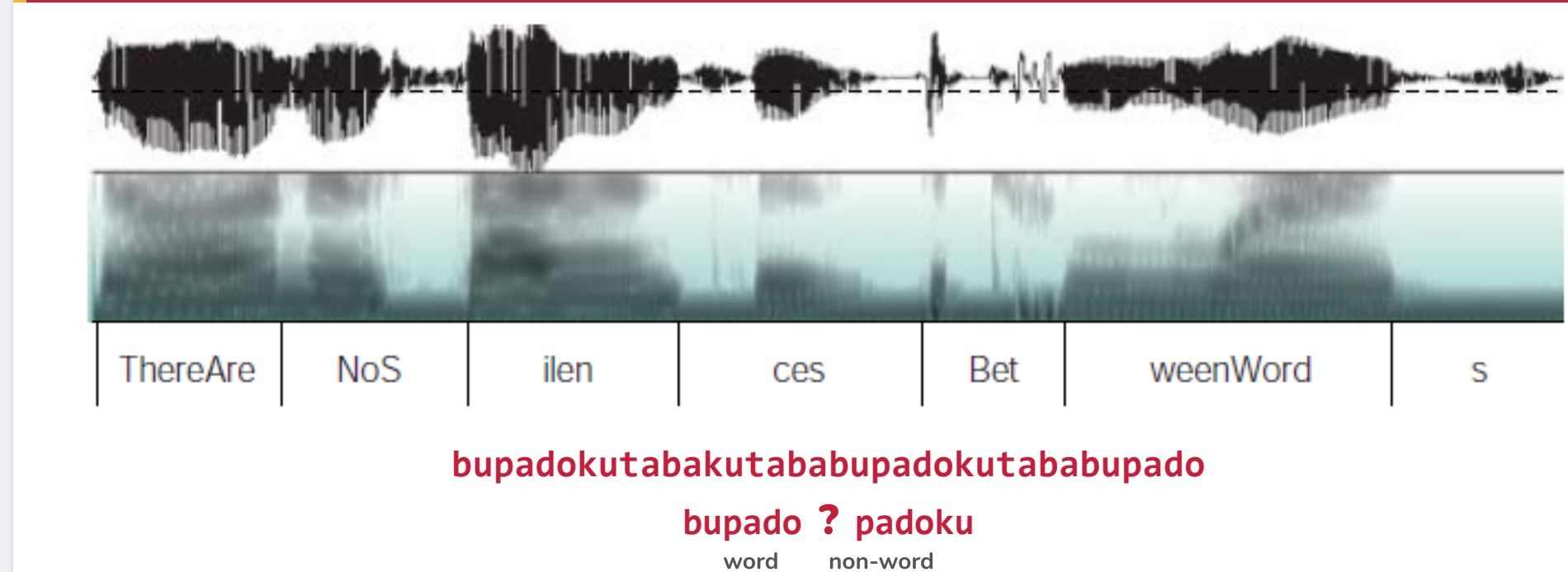


Inspiration · Statistical learning



Aims

- Is an RNN trained on next-token prediction able to learn words from an unsegmented stream?
- How is lexical information (word position, character, etc.) represented in a recurrent network that learns to produce words?
- How does the network's representation of information change with vocabulary complexity?
- How do the network's weights differ as a function of the complexity of the learned vocabulary (function → structure)?

Fig 1 · Mixed vocabularies & minimal DFAs

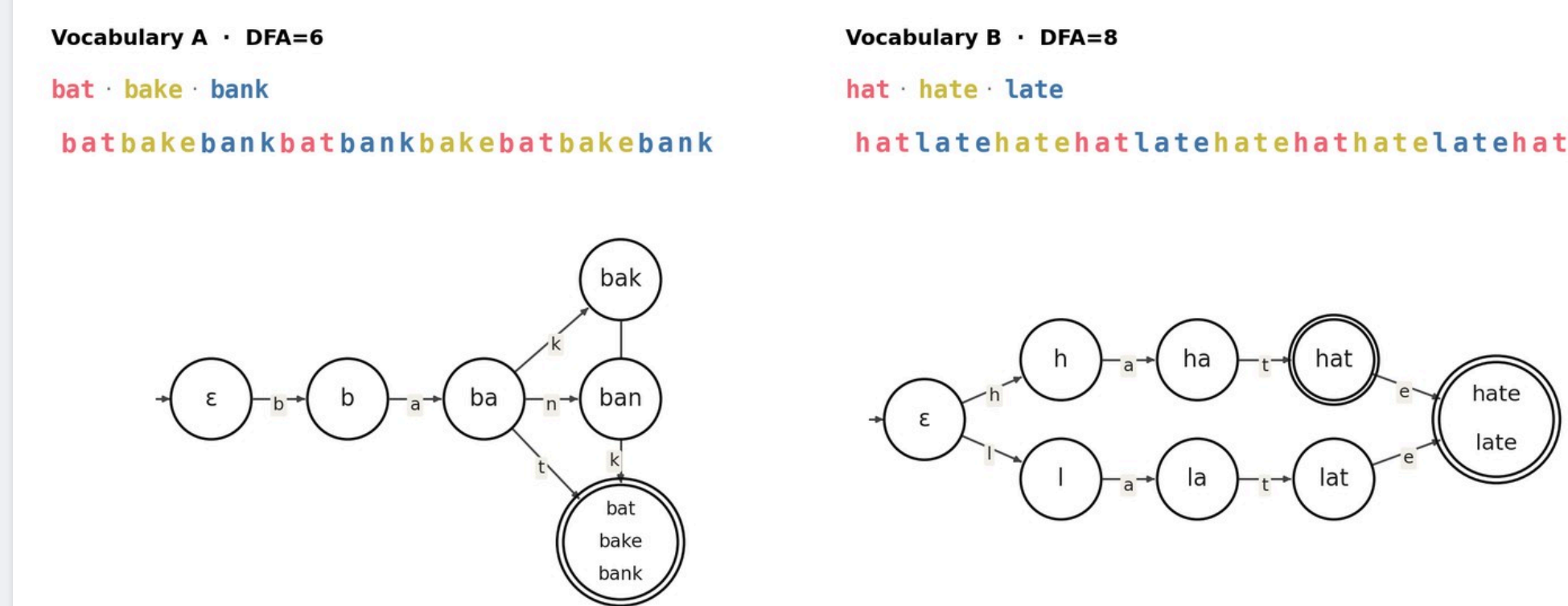


Fig 2 · Training the RNN

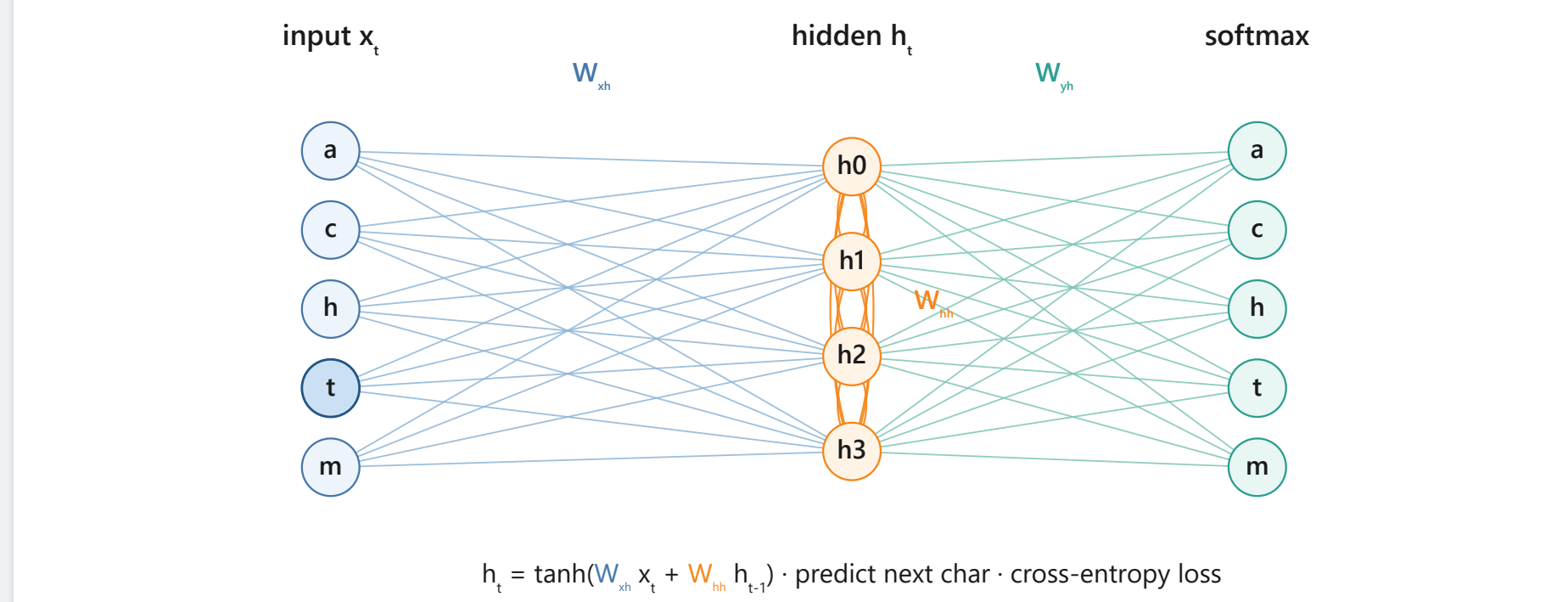


Fig 3 · Learning & generation

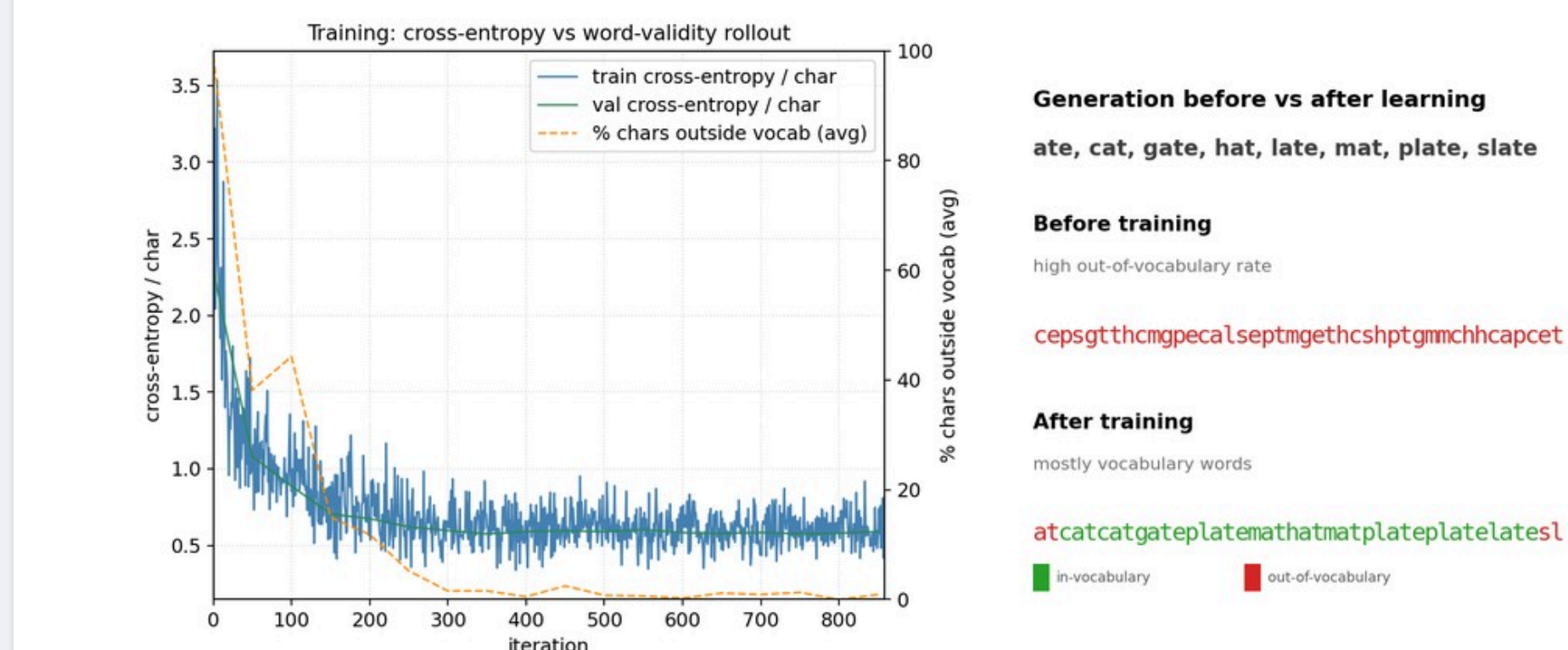


Fig 4 · Next-character probabilities

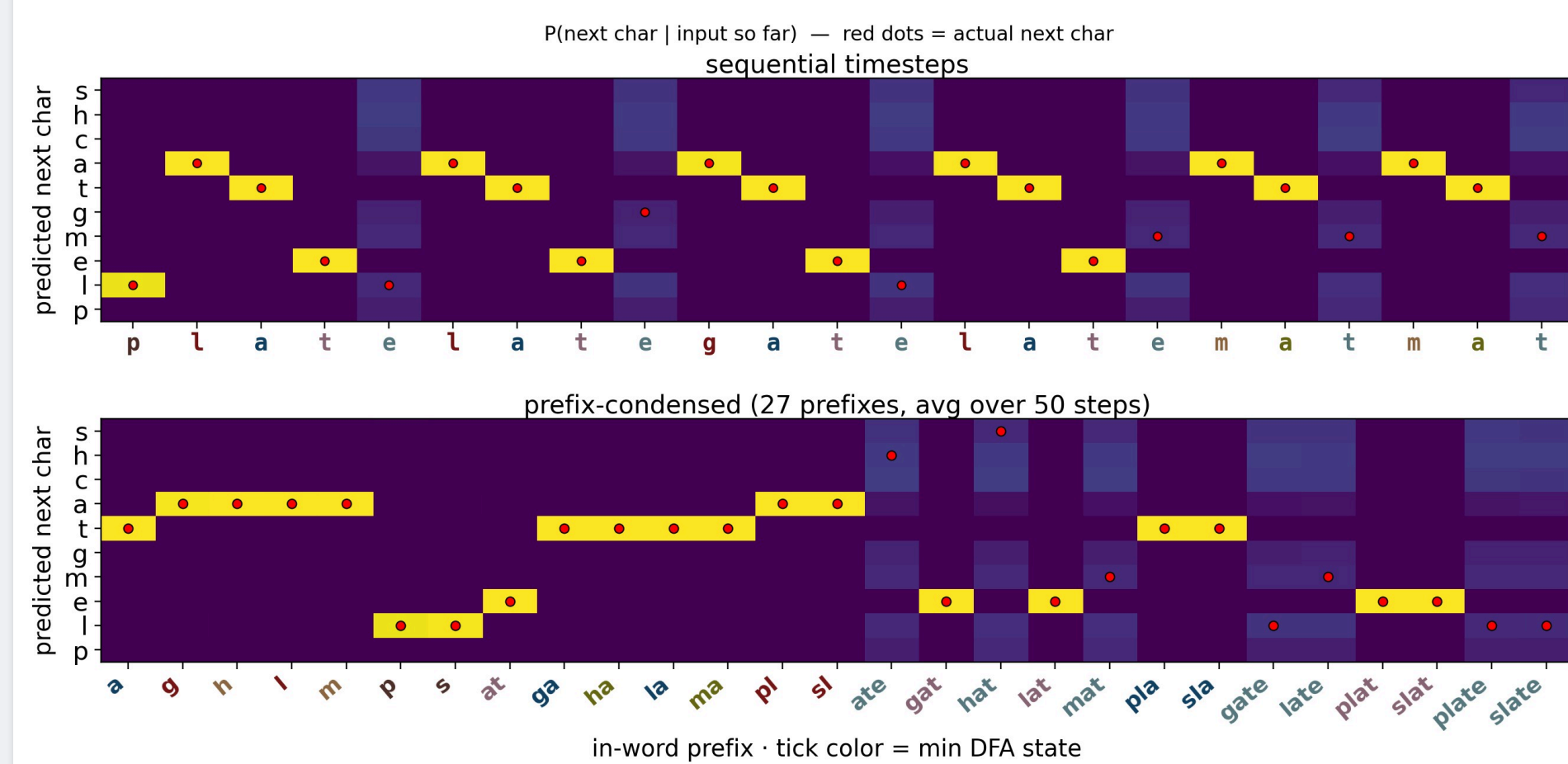


Fig 5 · Activation heatmap

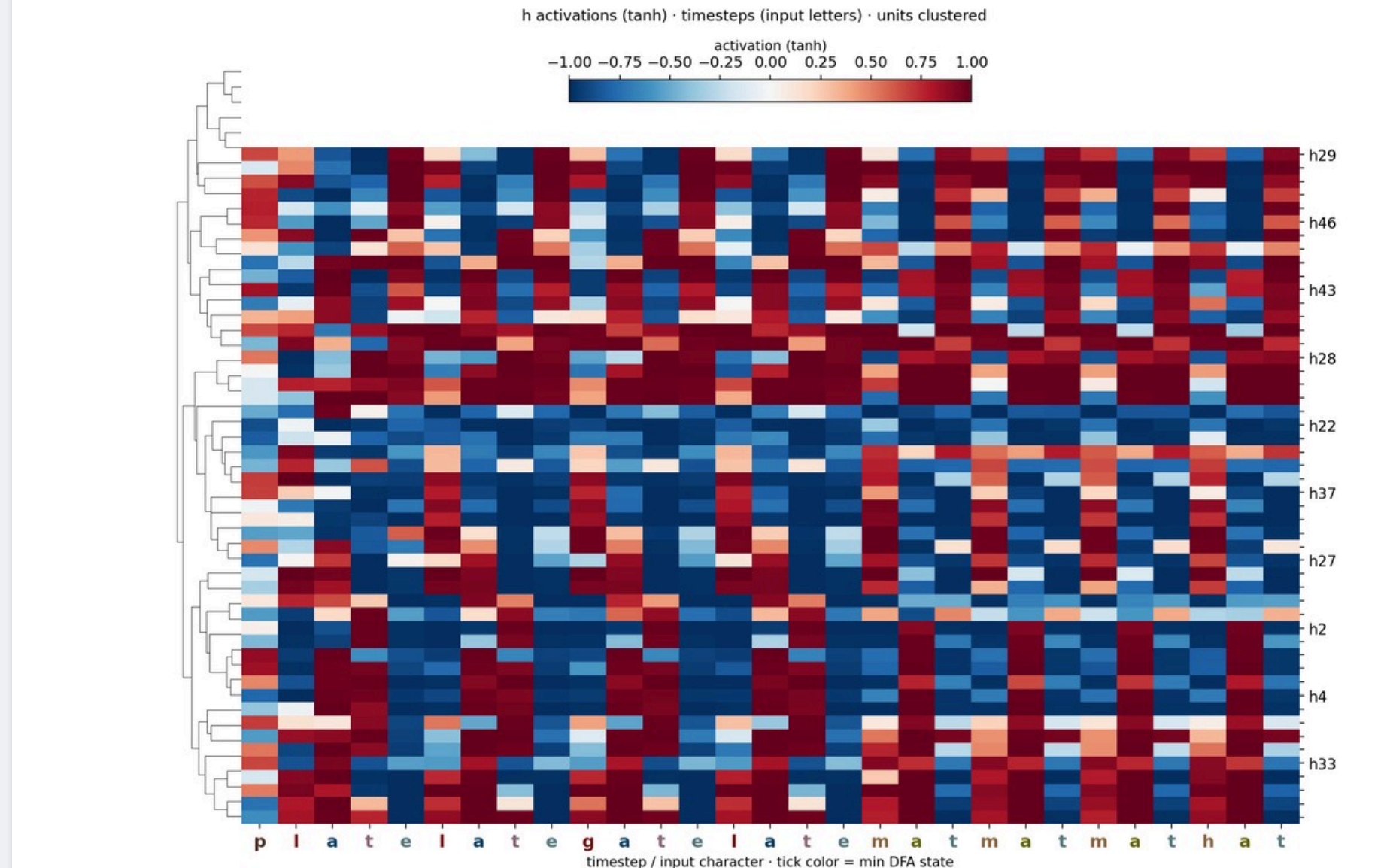


Fig 6 · Activations by prefix

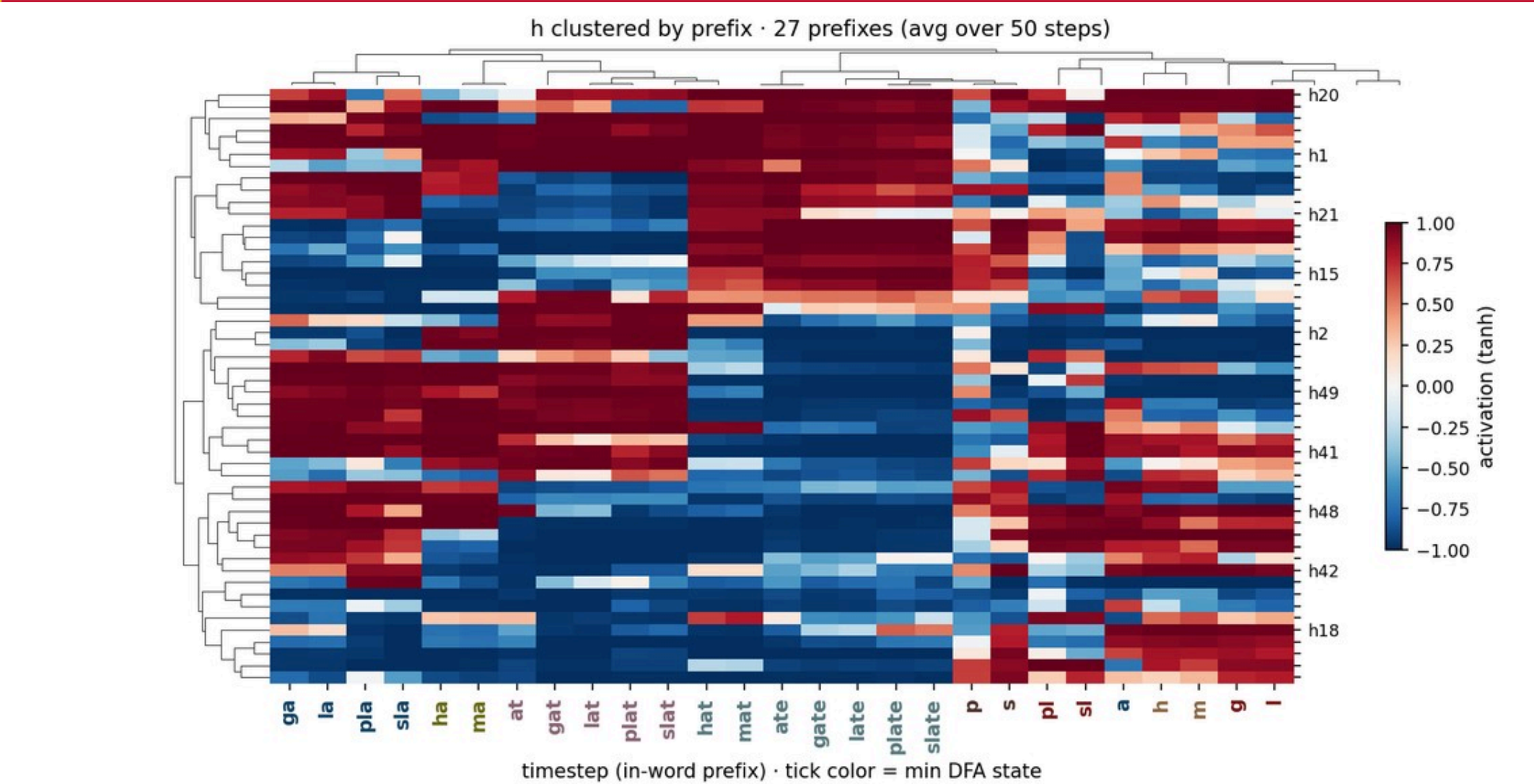


Fig 7 · Hidden-state correlation

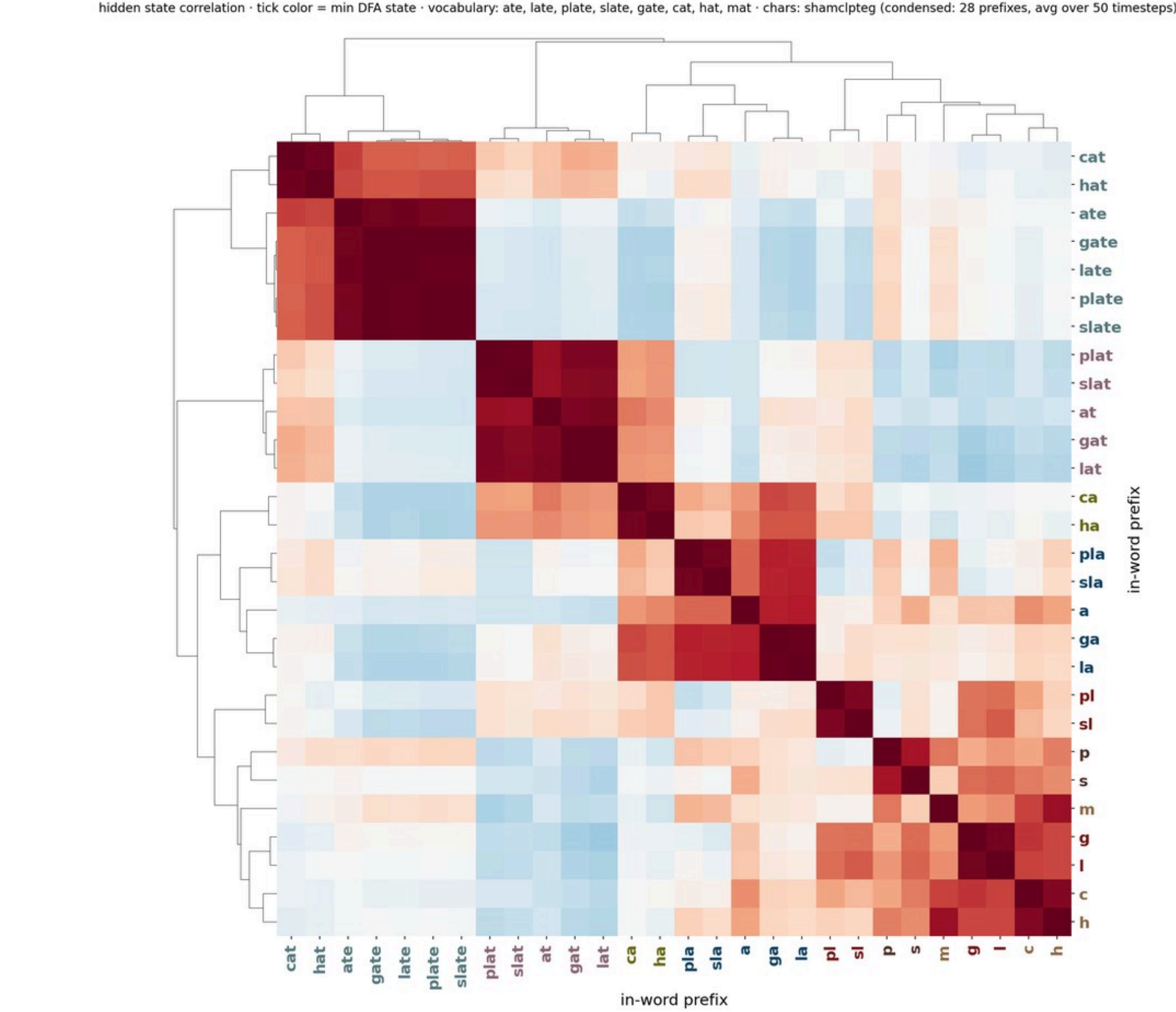


Fig 8 · DFA geometry & separation

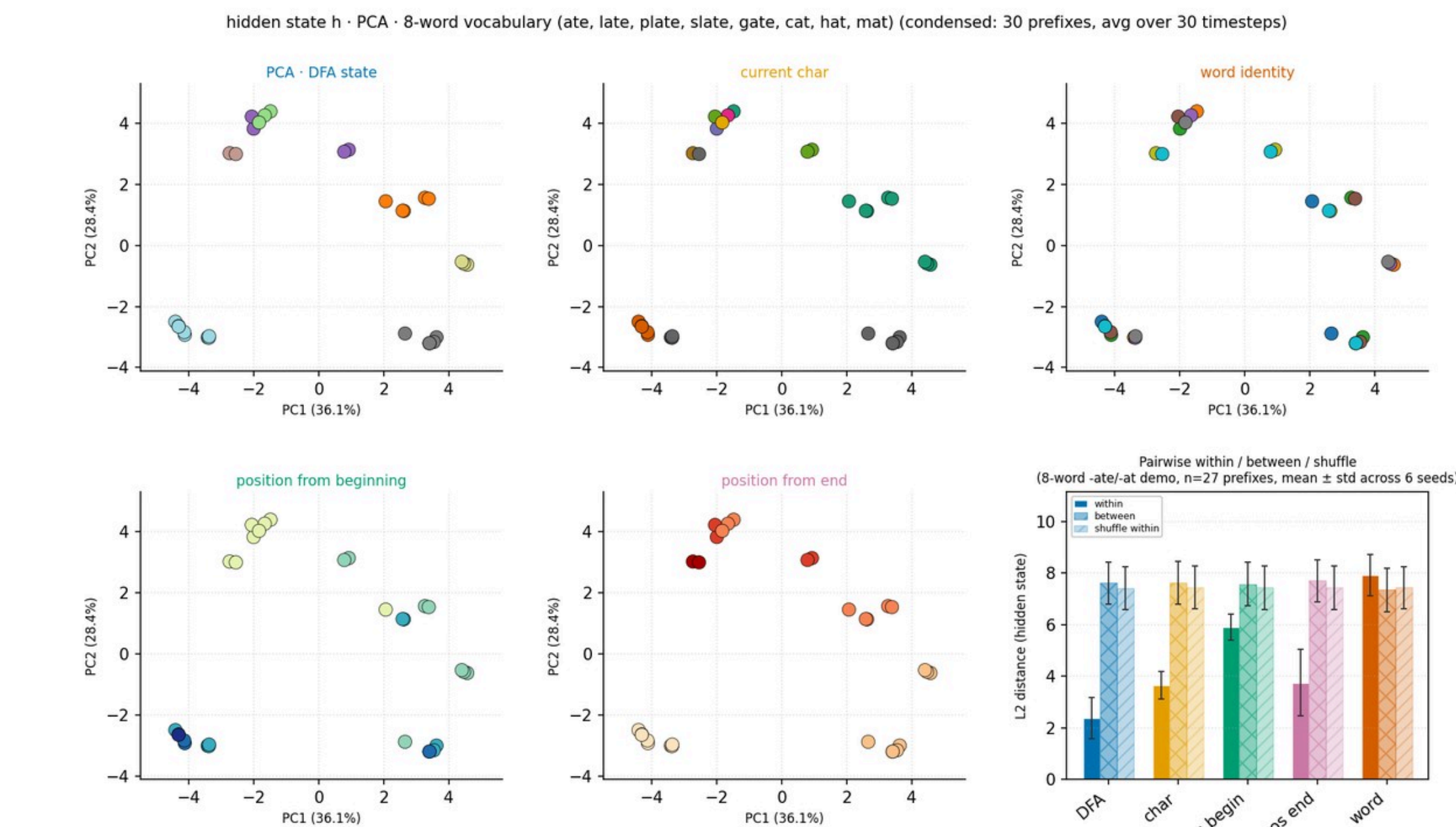


Fig 9 · Example selective units

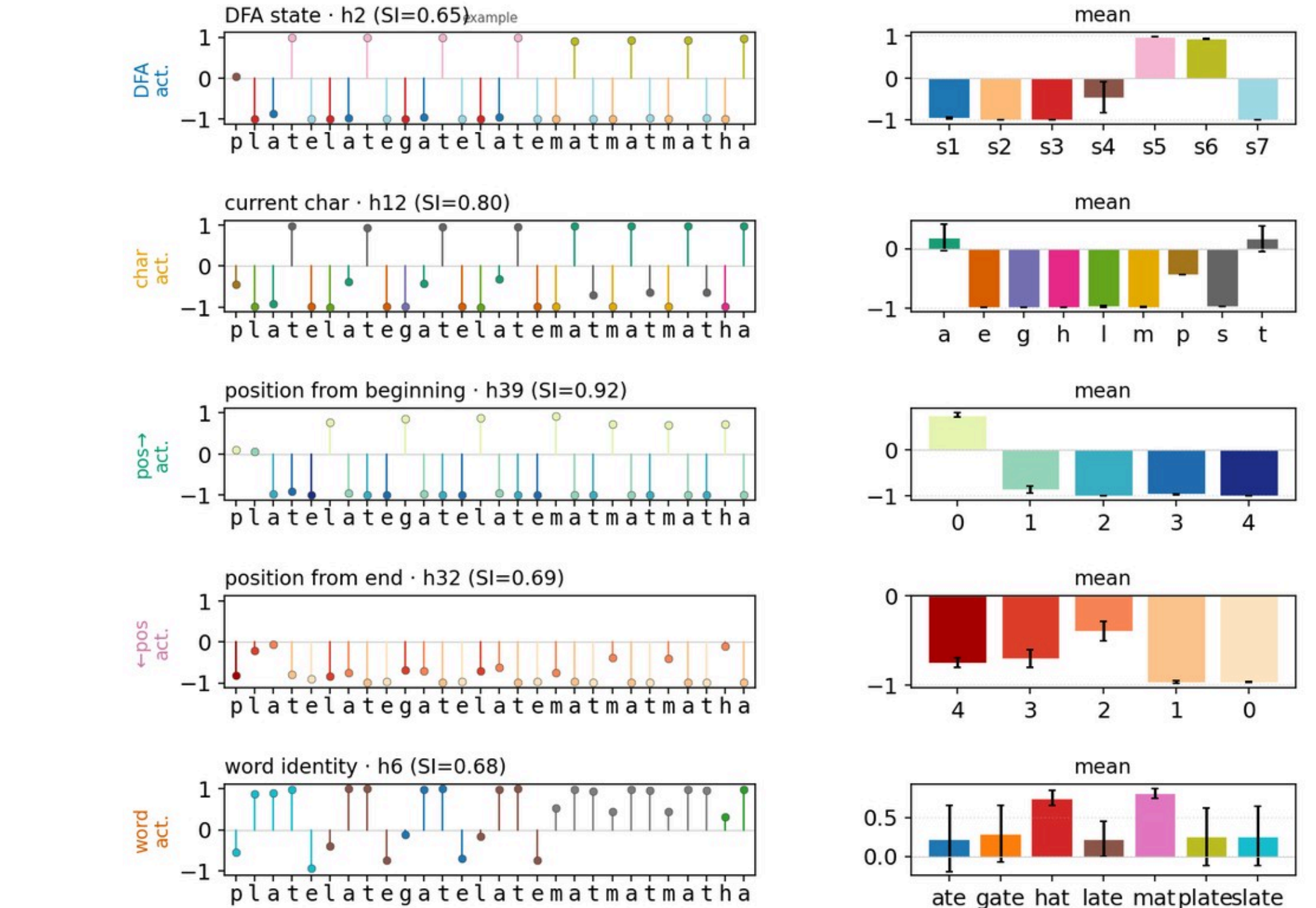


Fig 10 · Linear decoding

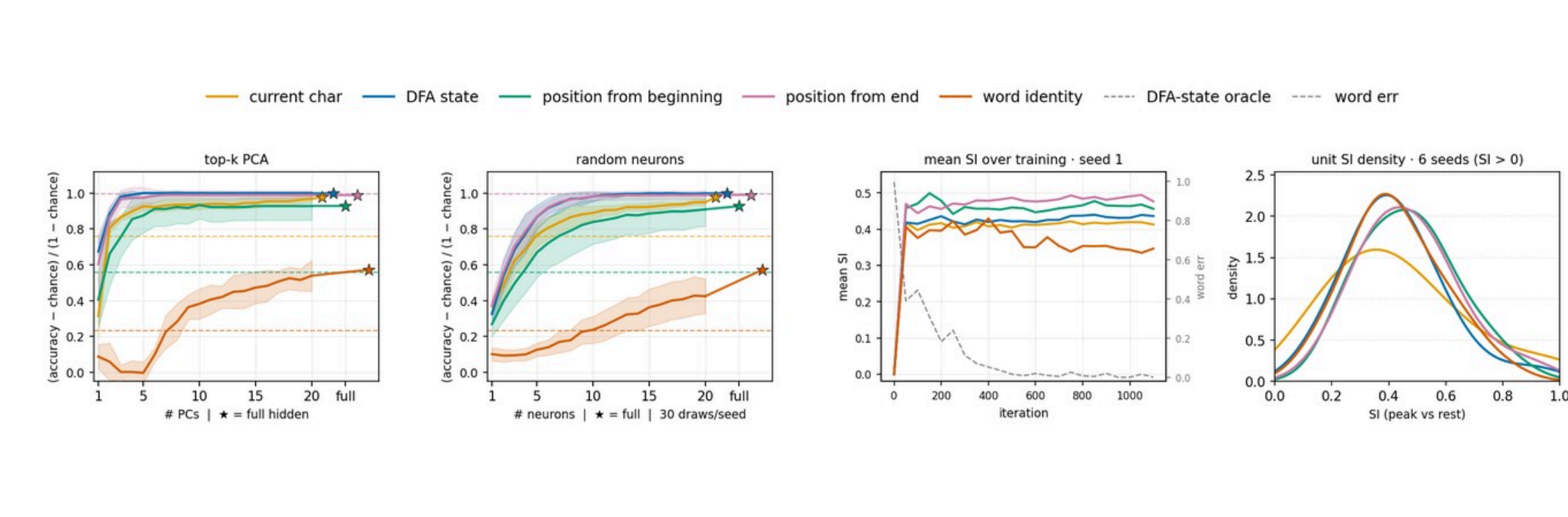


Fig 11 · Word trajectories

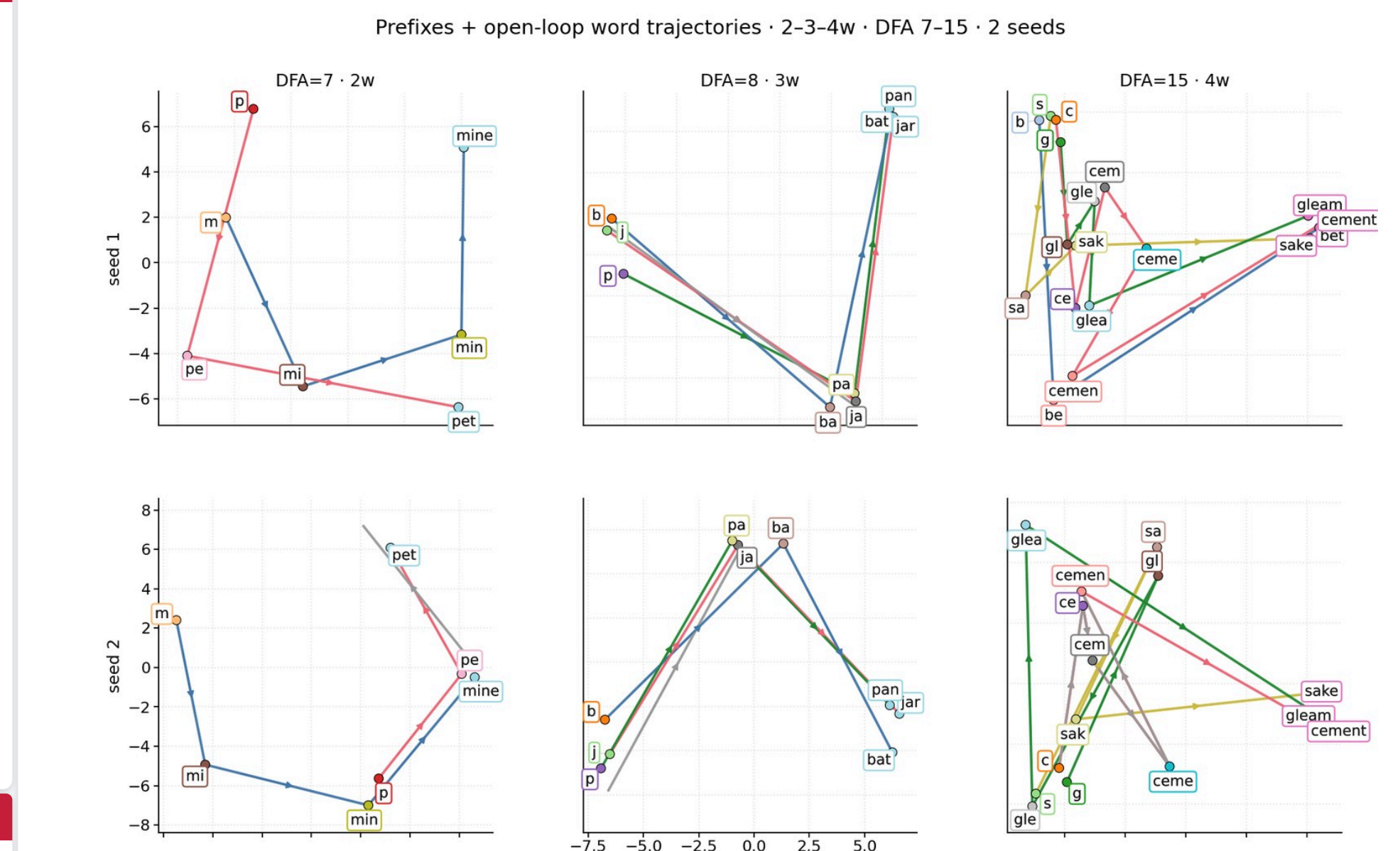


Fig 12 · Activations & scaling vs DFA size

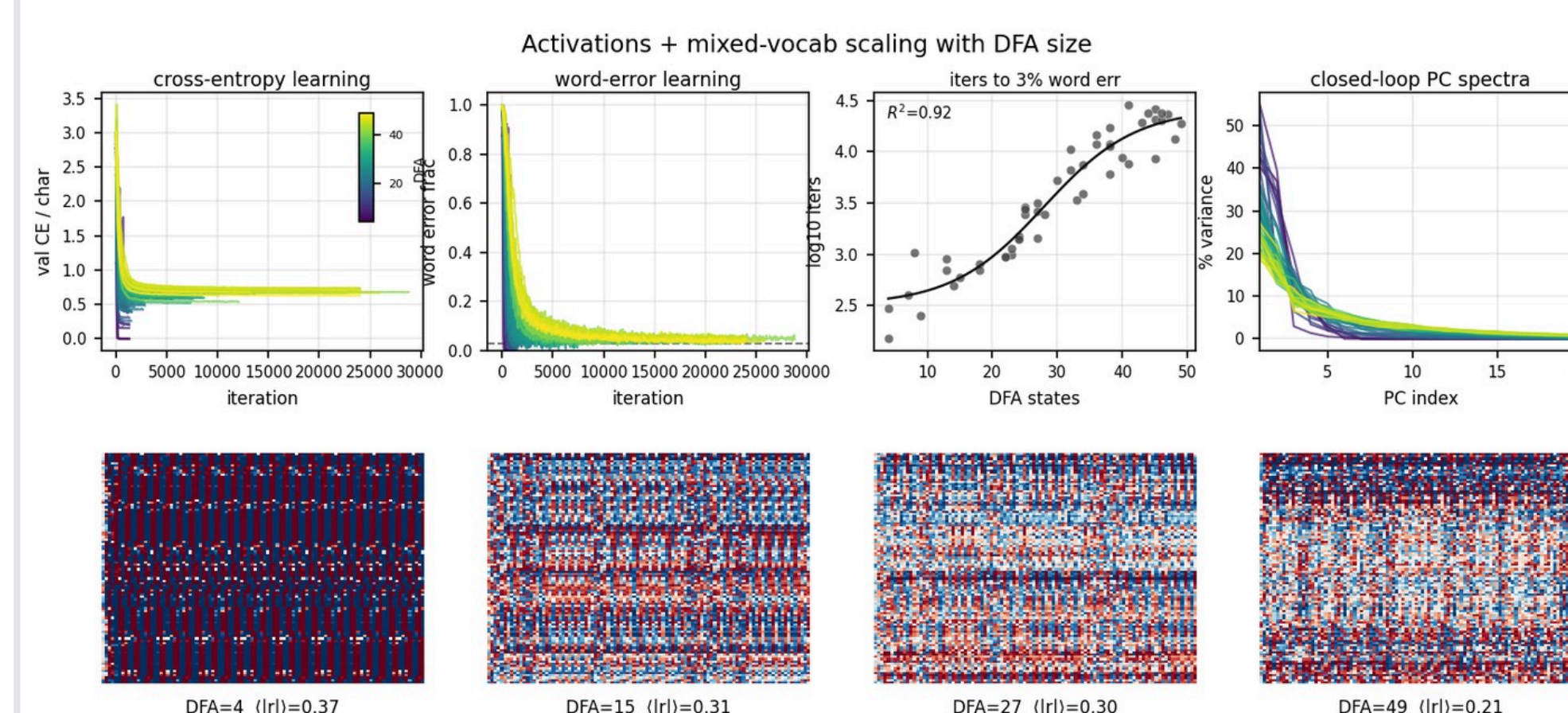


Fig 13 · Readouts vs DFA size

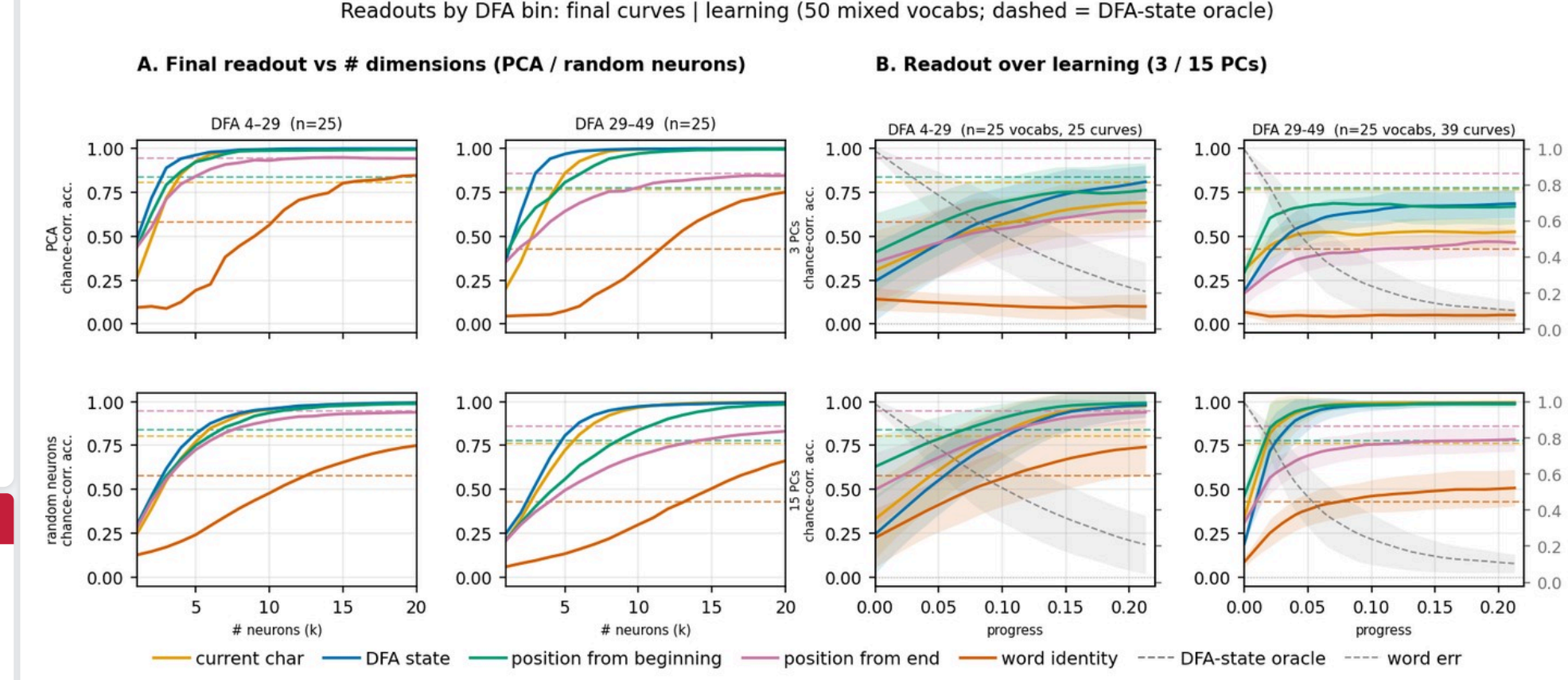


Fig 14 · Geometry & unit selectivity

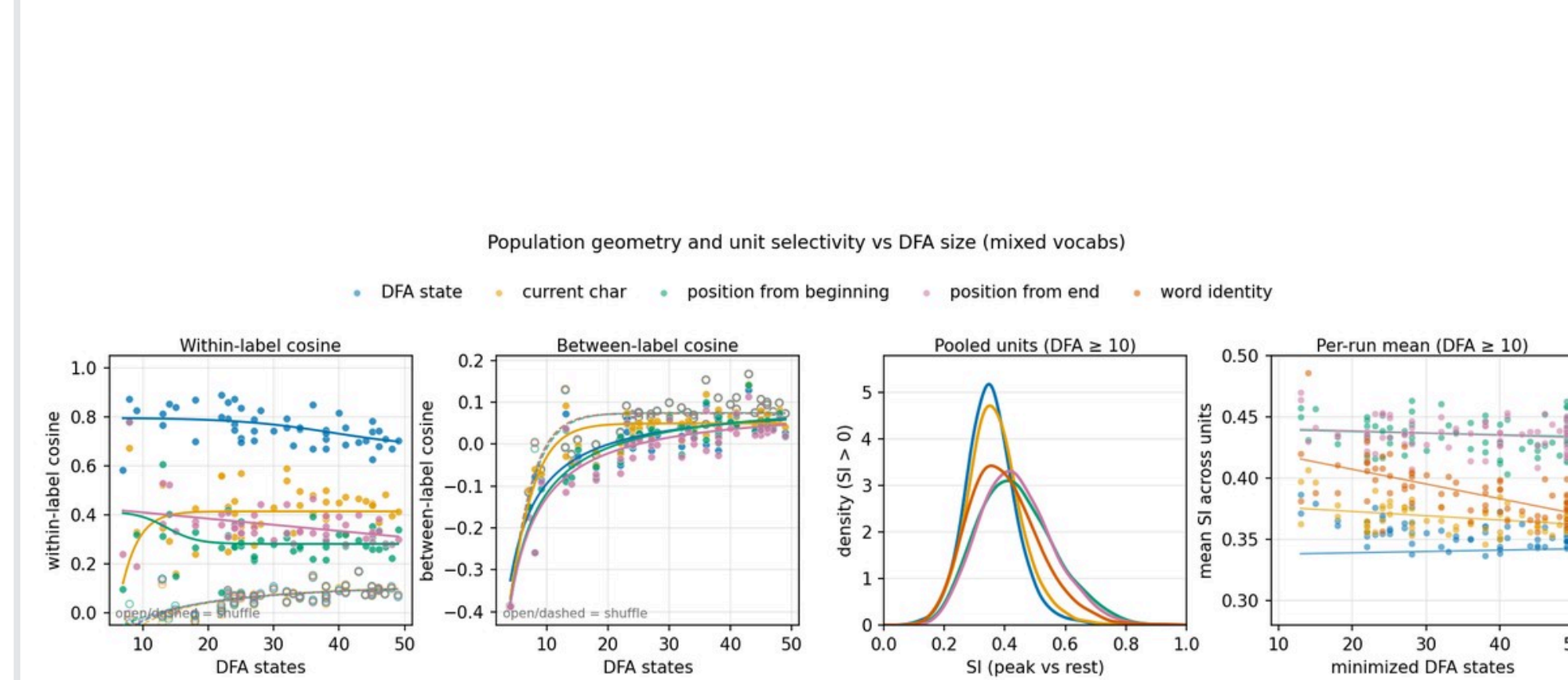


Fig 15 · Weight matrices vs DFA

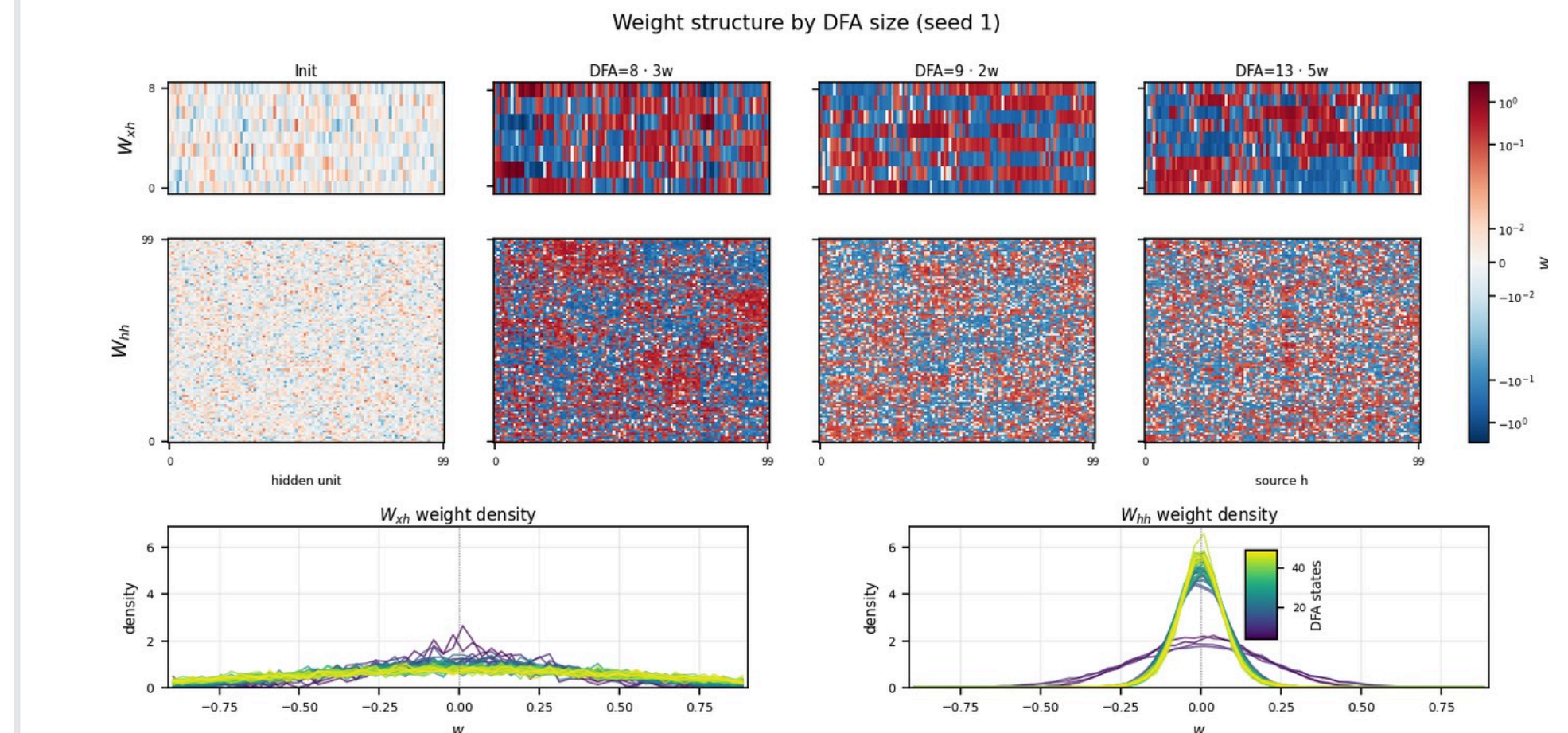


Fig 16 · Weight graph motifs vs DFA

